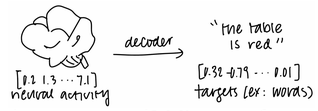
Task-driven Models for Neural Speech Representations

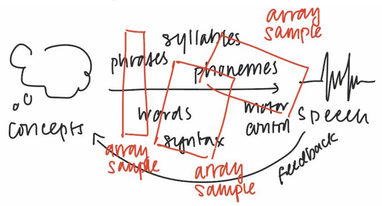
Yourui Shao, Alessandro Marin Vargas, Frank Willett, Neural Prosthetics Translational Lab, Stanford University

Background

Speech Representations for BCIs



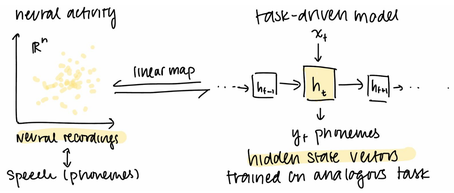
How is speech represented? Mixture of representations - how are different linguistic features represented together?



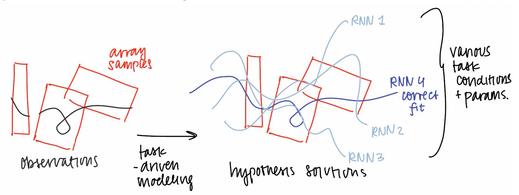
Problem: Neural recordings are small samples of all activity related to speech production

Task-Driven Modeling

RNNs produce computational analogs to neural activity [1,2]



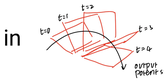
Mechanistically vary task conditions and parameters to test hypotheses on biological constraints & solutions [3,4]



1. Phoneme Sequencing

Question

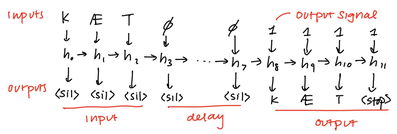
Phonemes are queued for output in overlapping subspaces.
Why are motions through subspaces not uniform over position/time?





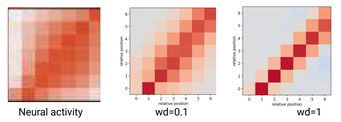
Methods

- Fineweb sentence dataset
- Learned to jointly encode phoneme at each position in hidden state
- Compare representations between timesteps



Tested:

1. hidden state dimension relative to sequence length
   - measured effective dimensionality
2. delay period lengths
3. task diversity (fwd/back)
4. max. vs. exact seq. lengths
5. linear vs. nonlinear solutions
6. **regularization**



Average cosine similarity between subspaces of phonemes at different positions, between adjacent time steps

Results

Uniform rotating queues emerge under regularization
- other parameters and task settings do not uniquely determine rotation depth

Regularization induces artificial NN solutions more similar to neural activity [5]
→ why is regularized solution uniform while neural activity is not?

Interpretation:
Differing conditions ⇒ neural representations likely don't receive phonemes serially

Methods

1. Train RNNs with task setting and parameters of interest (e.g., converting seq. of words to corresponding phonemes)
2. Run inference on test sentences and obtain hidden states
3. Train Ridge regression encoders and decoders on hidden states, obtaining feature representations
4. Analyze feature representations using magnitude, cosine similarity, encoding R^2, dimensionality

Linear Probes

Linear encoders: predict hidden state vector ($\mathbf{H}_t$) given phoneme, word, and syntax features to learn hidden state encodings. E.g., for one-hot encoded phoneme $\phi_{t,p}$ at sequence position p, time t

Single encoder:

$$\mathbf{H}_t = \mathbf{A}_{p,t}\phi_{p,t} + \boldsymbol{\epsilon}$$

Joint encoder:

$$\mathbf{H}_t = \boldsymbol{\epsilon} + \textstyle\sum_p \mathbf{A}_{p,t}\phi_{p,t}$$

Decoder (binary classification over target classes):

$$c_{\phi,p,t} = \mathbf{A}_{t,p}\mathbf{H}_t + \mathbf{b}_{t,p} + \boldsymbol{\epsilon} \in \{-1, 1\}$$

High encoding R^2, low decoding R^2 - feature is represented in hidden state, but in compressed / dimension reduced form

Discussion

Limitations

1. assumptions about word & phoneme input/output format → future work end-to-end (concept → phonemes)
2. assumptions made in features (what context is useful?)

Contributions

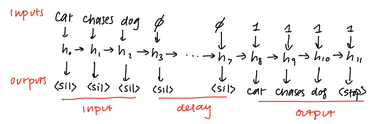
- analyzed learned solutions for speech production on phoneme sequencing, syntax, and word to phoneme levels
- method for adding contextual information to position-only sequence encoders for words and phonemes
- regularization as prerequisite for examining mechanistic effects on learned representations
  - dimensionality reduction explanation

2. Syntax Representations

Question

Prelim. work finds neural representations with tuning for syntax.
Does regularization induce syntactic representations? Why?

Methods



Artificial dataset - sentence templates



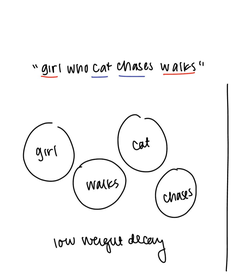
Single feature encoders & decoders for
- word at position
- sentence structure
- part of speech at position
- number on subj/verb pairs

Delay period hidden states
Vary regularization, ungrammatical controls

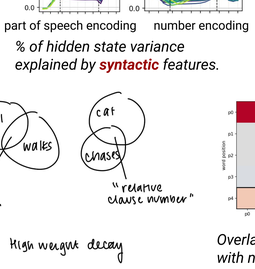
Results

- Weight decay => more encoding for syntax
- Syntax equivalent to & makes possible overlaps in word representations
- Linguistic syntax ⇔ compressive word representations (co-currence)

We theorize weight decay induces syntax as L2 regularization can cause dimensionality reduction [6]

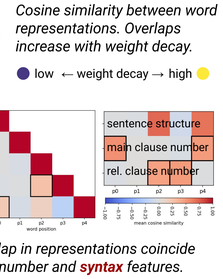


Sentence structure *encoding and decoding R^2, averaged across seeds.*



Word encoding and decoding R^2 (word at position 4).

% of hidden state variance explained by ***syntactic*** *features.*

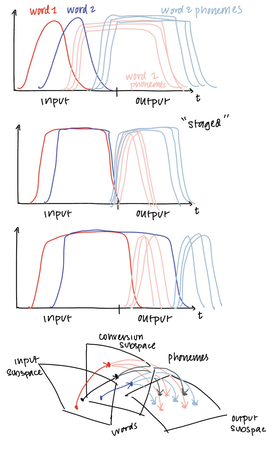


Cosine similarity between word representations. Overlaps increase with weight decay.

low ← weight decay → high

Overlap in representations coincide with number and ***syntax*** *features.*

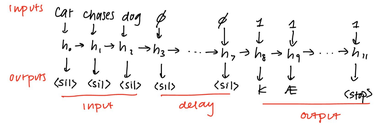
3. Word to Phoneme Translation



Question

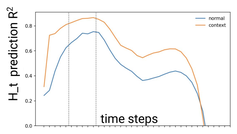
Serial phoneme input unlikely – how are words converted into phonemes?

Methods



Jointly encode words and phonemes, with **contextual features**
- phoneme at each position
- **phoneme by position of word parent**
- word at each position
- **word by time relative to phoneme output**

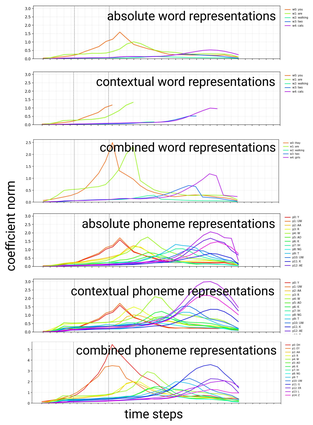
"we are feeding **two** cats."
W IY | AA R | F IY D IH NG | T **UW** | K AE T S
"we do not see **you**."
W IY | D UW | N AA T | S IY | Y **UW**



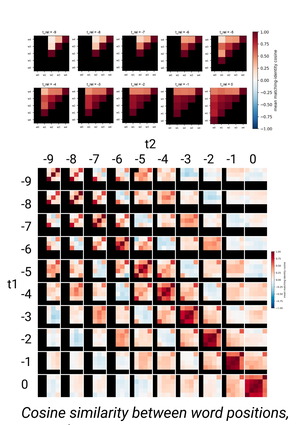
Top: Phonemes UW in the same position, but belonging to different parent words, likely represented differently over t. Bottom: % of hidden state explained by features, normal (blue) vs. ***contextual*** *(orange).*

Results

- phonemes not recoverable until time of output => word to phoneme conversion during output
- evidence of word -> phoneme conversion subspace



*Phoneme and word representation magnitudes (****contextual*** *vs. non-contextual)*



Cosine similarity between word positions, time relative to output

References

1. Mante, V., Sussillo, D., Shenoy, K. V. & Newsome, W. T. Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature* **503**, 78–84 (2013).
2. Yang, G. R. et al. Task representations in neural networks trained to perform many cognitive tasks. *Nat. Neurosci.* **22**, 297–306 (2019).
3. Marin Vargas, A. et al. Task-driven neural network models predict neural dynamics of proprioception. *Cell* **187**, 1745–1761.e19 (2024).
4. Driscoll, L. N., Shenoy, K. & Sussillo, D. Flexible multitask computation in recurrent networks utilizes shared dynamical motifs. *Nat. Neurosci.* **27**, 1349–1363 (2024).
5. Sussillo, D., Churchland, M. M., Kaufman, M. T. & Shenoy, K. V. A neural network that finds a naturalistic solution for the production of muscle activity. *Nat. Neurosci.* **18**, 1025–1033 (2015).
6. Galanti, T., Siegel, Z. S., Gupte, A. & Poggio, T. A. SGD with weight decay secretly minimizes the ranks of your neural networks. In *Conference on Parsimony and Learning* **280**, 1388–1412 (PMLR, 2025).